Acid–Base Properties of Salts

Hydrolysis, Ka × Kb = Kw and salt pH

Name _____ Class _____ Date _____

DATA AT 25 °C

$K_w = 1.0 \times 10^{-14}$ · $K_a \times K_b = K_w$ · pH + pOH = 14.00

Ka: CH_3COOH 1.8×10^{-5} · HF 6.8×10^{-4} · HCN 4.9×10^{-10} · Kb: NH_3 1.8×10^{-5}

For a weak acid or base: $x \approx \sqrt{(K \times C)}$, if x is under 5 % of C

Write the hydrolysis equation for the ion that reacts. Check the 5 % rule.

A. Predicting

1. Predict whether each solution is acidic, basic or neutral, and name the ion responsible: (a) $NaNO_3$ (b) K_2CO_3 (c) NH_4Br (d) CH_3COOK (e) LiCl (f) NH_4NO_3

2. Write the equation for the reaction with water of (a) the fluoride ion (b) the ammonium ion.

B. Ka and Kb

3. Calculate: (a) Kb of F^- (b) Ka of NH_4^+ (c) Kb of CN^-. Which is the strongest base, F^- or CN^-?

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C. pH of salt solutions

4. Calculate the pH of 0.10 mol/L NaCN.

5. Calculate the pH of 0.050 mol/L NH_4NO_3.

6. Calculate the pH of 0.25 mol/L potassium ethanoate, CH_3COOK.

D. Titrations

7. 25.00 mL of 0.100 mol/L ethanoic acid is titrated with 0.100 mol/L NaOH. (a) What volume of NaOH reaches the equivalence point? (b) What is the concentration of ethanoate at that point? (c) Calculate the pH at the equivalence point and suggest a suitable indicator.

E. Challenge

8. Explain why a solution of aluminium chloride, $AlCl_3$, is acidic even though Cl^- is neutral.

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